Status	Finished
Started	Sunday, 18 October 2026, 9:14 AM
Completed	Sunday, 18 October 2026, 9:20 AM
Duration	5 mins 21 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is large and sorted
- c. When the list is large and unsorted
- d. When the list is small and sorted

Question **2**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The first element is returned
- b. The last element is returned
- c. The search is considered unsuccessful ✓
- d. An error is raised

Question **3**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It compares elements sequentially
b. It always starts from the beginning of the list
c. It can be applied only if the list is sorted ✓
d. It works on unsorted lists

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first
b. By using a hash function
c. By comparing each element in the list sequentially ✓
d. By dividing the list into halves

Question **5**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Sorting elements in a list
b. Inserting elements into a list
c. Deleting elements from a list
d. Determining whether an element is present in a list ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Sorting the list before searching
b. Dividing the list in half repeatedly
c. Checking each element sequentially ✓
d. Skipping every second element

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓
b. Linear search divides the list into halves.
c. Linear search is more efficient than binary search on large lists.
d. Linear search requires the list to be sorted.

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers
b. It is slower than linear search for small lists
c. It requires the list to be sorted ✓
d. It can only be applied to large lists

Question **9**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Binary search
c. Linear search ✓
d. Jump search

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Merge search
b. Linear search ✓
c. Quick search
d. Bubble search

Question **11**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element
b. Binary search has a lower time complexity on large, sorted lists ✓
c. Binary search works on unsorted lists
d. Binary search does not require dividing the list

Question **12**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(1) ✓
- b. O👎
- c. O(log n)
- d. O(n log n)

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is sorted ✓
- b. When the list is unsorted
- c. When the list is very small
- d. When the list contains duplicate elements

Question **14**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 90 and 99 ✓
- b. 89 and 94
- c. 90 and 100
- d. 94 and 99

Question **15**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array

b. When the item is somewhere in the middle of the array ✓

c. Item is the last element in the array or item is not there at all

d. When the item is not the array at all